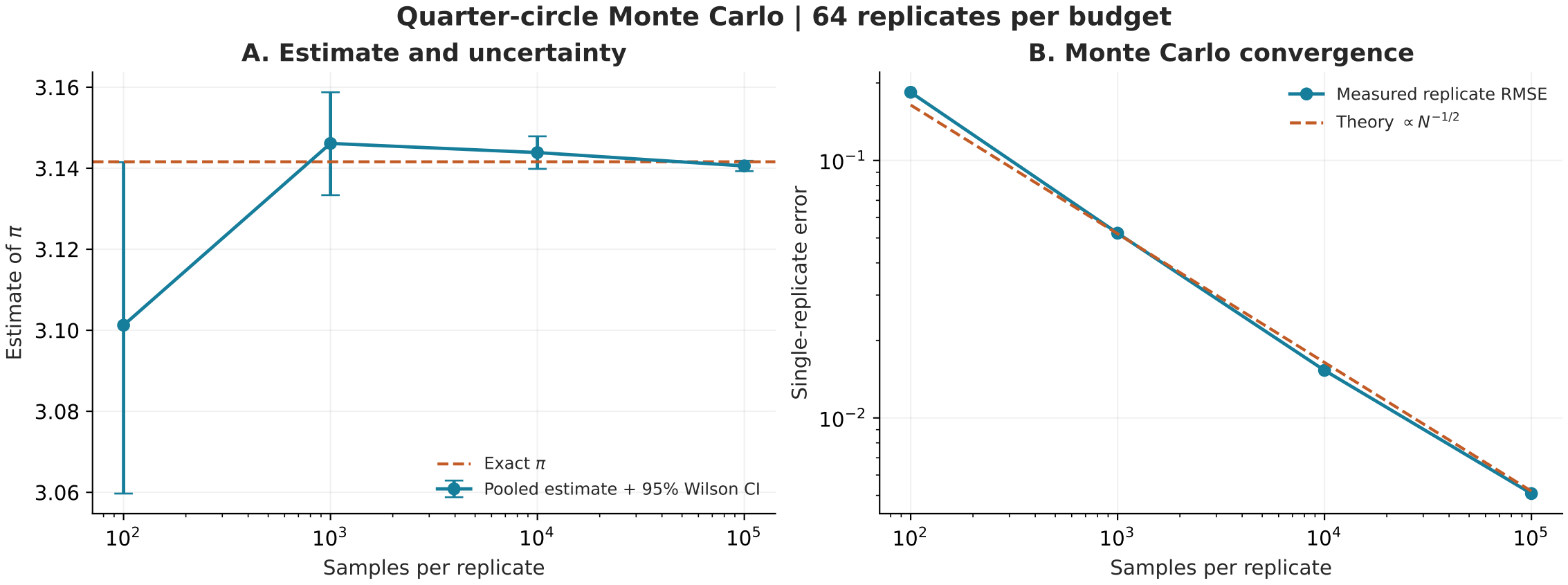
Quarter-circle Monte Carlo | 64 replicates per budget
A. Estimate and uncertainty
B. Monte Carlo convergence
Estimate of π
3.16
3.14
3.12
3.10
3.08
3.06
Exact π
Pooled estimate + 95% Wilson CI
10^2
10^3
10^4
10^5
Samples per replicate
Single-replicate error
10^-1
10^-2
Measured replicate RMSE
Theory ∝ N^-1/2
Samples per replicate